Raphatelo Mosima Dorothy Activity1.2(Problem 4)

1. IPO

INPUT	PROCESSING	OUTPUT
FirstNumber SecondNumber	Get FirstNumber Get SecondNumber Calculate HighestCommonFactor	HighestCommonFactor

2. ALGORITHM

START

INPUT FirstNumber

INPUT SecondNumber

IF FirstNumber < SecondNumber

Then

Factor = FirstNumber

Else

Factor = SecondNumber

ENDIF

WHILE (FirstNumber MOD Factor != 0)

OR (sECONDNumber MOD Factor != 0)

Factor = Factor – 1

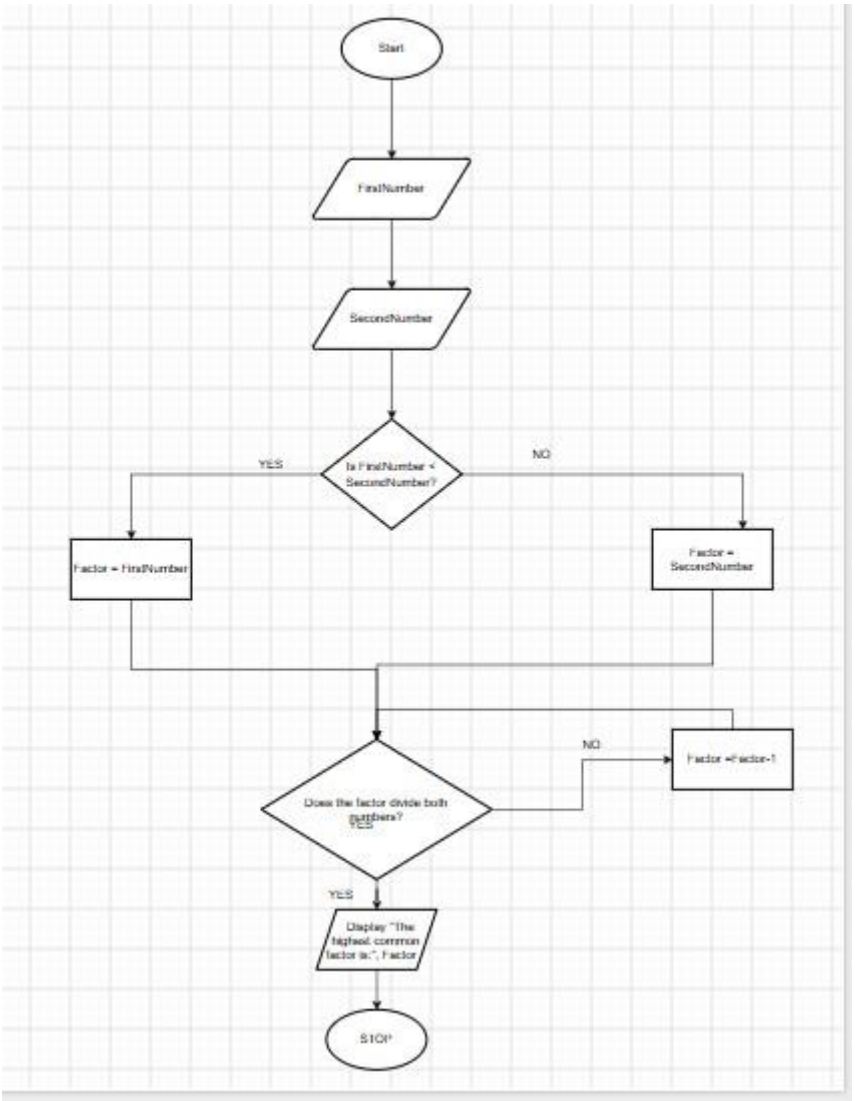
ENDWHILE

DISPLAY “The highest common factor is: “, Factor

STOP

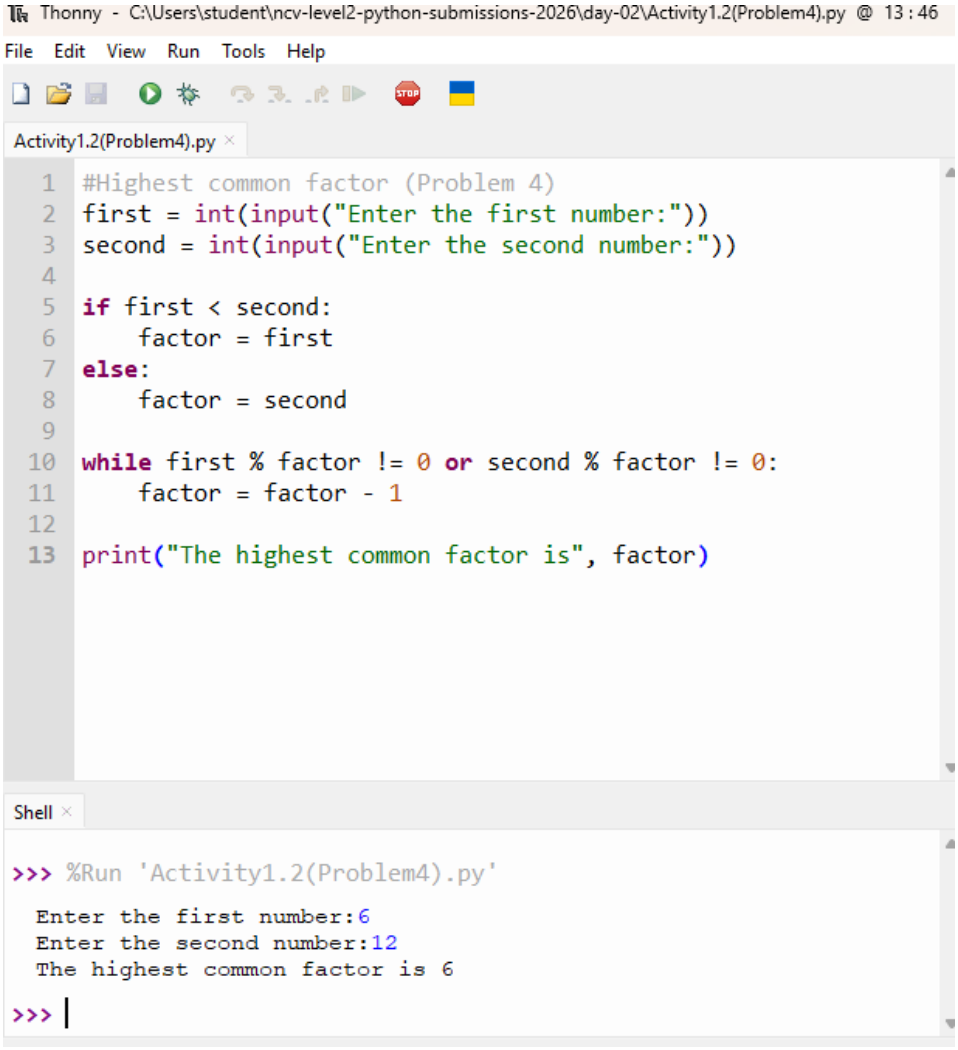
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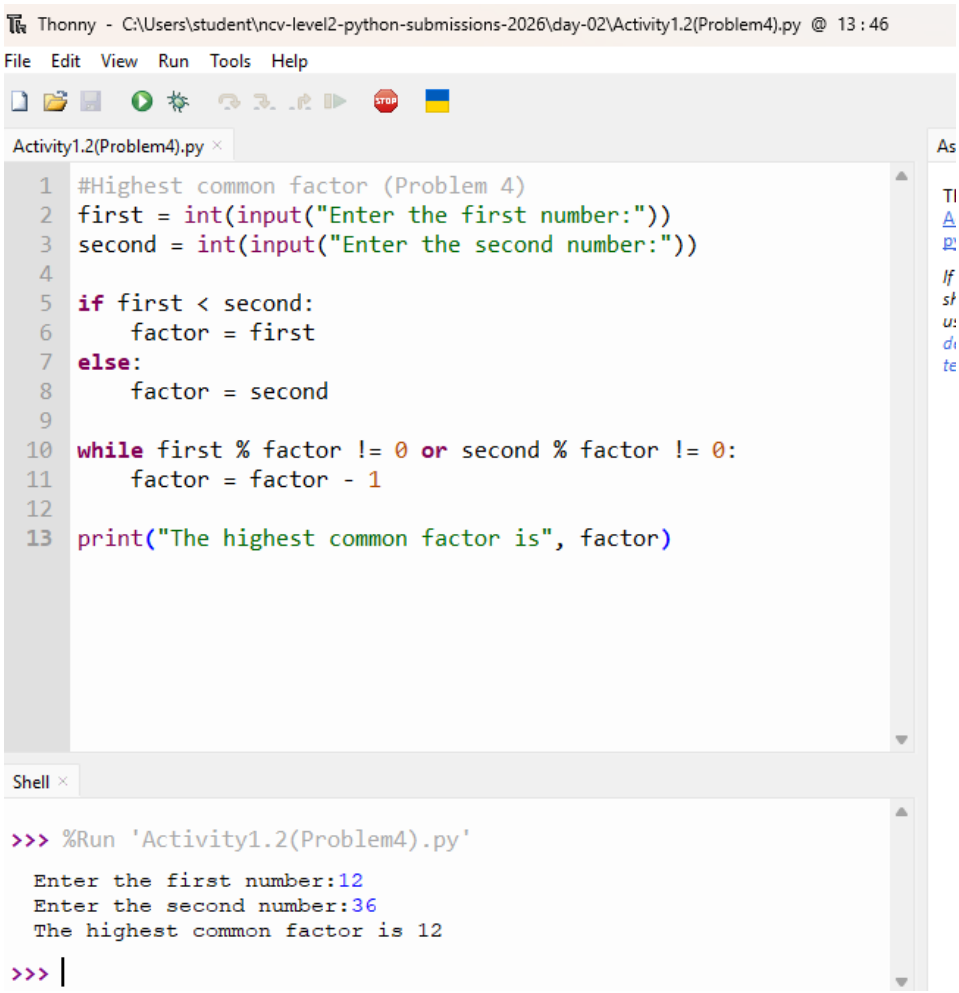
3. FLOWCHART



3. TRACE TABLE

Test case	FirstNumber	SecondNumber	Expected Results
1	6	12	6
2	12	36	12
3	12	18	6




Thonny - C:\Users\student\ncv-level2-python-submissions-2026\day-02\Activity1.2(Problem4).py @ 13 : 46
File Edit View Run Tools Help
Activity1.2(Problem4).py
1 #Highest common factor (Problem 4)
2 first = int(input("Enter the first number:"))
3 second = int(input("Enter the second number:"))
4
5 if first < second:
6 factor = first
7 else:
8 factor = second
9
10 while first % factor != 0 or second % factor != 0:
11 factor = factor - 1
12
13 print("The highest common factor is", factor)
Shell
>>> %Run 'Activity1.2(Problem4).py'
Enter the first number:12
Enter the second number:36
The highest common factor is 12
>>>